SIMATS ENGINEERING

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ASSIGNMENT - 1.

TASK-1 - THREE CONCEPTUAL DESIGNS.

	A: Dated. relationship	B: Loan as entity (Preferred).	C. Ternary. LENDER - CURATOR - EXHIBITION.
Identity	No own id: key = lender + artifact + date.	Own key loan_id: LOAN_OBJECT.	No loan: the triple is the fact.
Degree	Binary	Binary + association entity.	Ternary.
History.	One date pair per pair	Many loans, Many artifacts, dates on LOAN / LOAN-OBJECT.	Dates on triple only.
Condition evidence.	Cannot attach. (No-loan row)	Owned by LOAN-OBJECT. One report per object per loan stage)	Attached to exhibition - wrong.
Failure & fixes.	- Calls: same artifact twice same date: many artifacts per loan (All possible)	Reports filed by curator name : expiry not linked to slot.	Shows a ternary can create false facts when apply (Task-3).

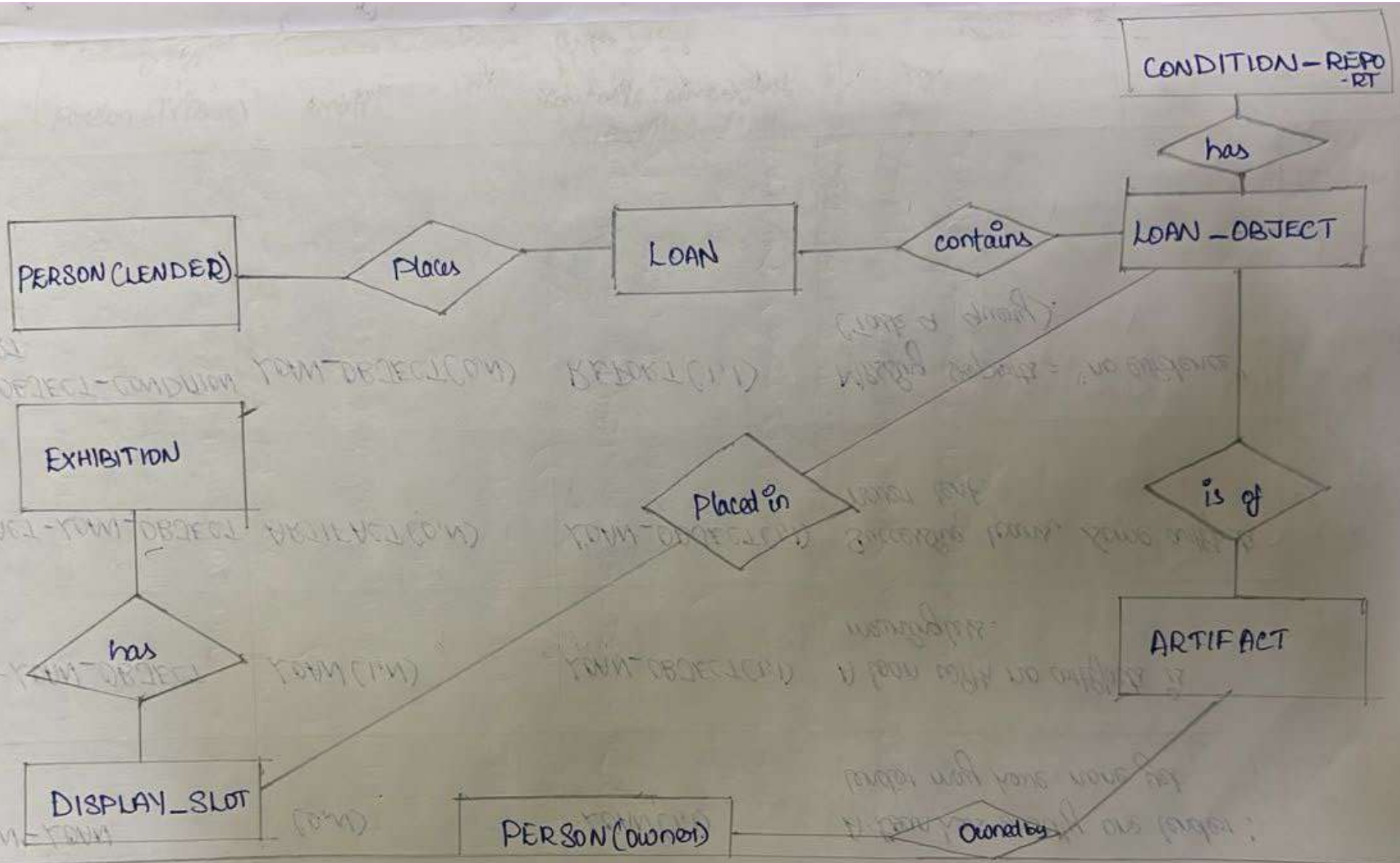
TASK-2- Entities, Attributes, Keys.

Entity	Primary Key	Candidate/other keys	Type notes
PERSON	Person_Id (Auto)	email	Composite: name→first, last, multivalued: phone: Contact = Category
ARTIFACT	artifact_Id	Catalog_no (unique)	Collection type → Its own entity (has data); value is on artifact
LOAN	loan_Id	loan_ref	Derived: loan_days = end-start (don't store)
LOAN_OBJECT	(loan_Id, artifact_Id)	–	Stores: one-time valuation, condition_in/out
CONDITION_REPORT	report_Id	(loan_Id, artifact_Id, report_date)	Evidence entity
EXHIBITION	exhibition_Id	–	name NOT a key
DISPLAY_SLOT (weak)	(exhibition_Id, slot_no)	–	Existence-dependent on EXHIBITION

TASK-3- RELATIONSHIPS AND CONSTRAINS

Relationship.	Lender Side (min, max)	Other Side (min, max)	Why
PERSON-LOAN	(0,N)	LOAN (1,1)	A Loan has exactly one lender; lender may have none yet.
LOAN-LOAN_OBJECT	LOAN (1,N)	LOAN_OBJECT (1,1)	A loan with no artifacts is meaningless.
ARTIFACT-LOAN_OBJECT	ARTIFACT (0,N)	LOAN_OBJECT (1,1)	Successful loans, some artifacts never lent.
LOAN_OBJECT-CONDITION_REPORT	LOAN_OBJECT (0,N)	REPORT (1,1)	Missing reports = "no evidence" (Task 9 query).

TASK-4 - ER DIAGRAM.



TASK-5- EER.

- GENERALIZATION- PERSON (supertype) → LENDER, EMPLOYEE; EMPLOYEE → CURATOR, REGISTRAR.
- CONSTRAINS: LENDER/EMPLOYEE = overlapping (a curator can lend a private object); CURATOR/REGISTRAR = overlapping too; employee specialization = partial (guards, admin staff have no subtype).
- DISCRIMINATOR: employee.Role ('CURATOR'/REGISTRAR') decides which subtype row must exist. A curator-lender has the same person_id in CURATOR and LENDER, so custody and condition records point to one identity.
- UNION (category): NOTIFICATION_CONTACT = UNION of PERSON and ORGANIZATION (one or the other)
- Subtypes vs one wide record: wide record has may NULLS and cannot show overlap rules; Subtypes keep only relevant fields.

TASK-6- WEAK ENTITIES, COMPLEX ATTRIBUTES.

- DISPLAY_SLOT: owner EXHIBITION, local (partial) key slot_no, full key (exhibition_id, slot_no). Identifying link = part of the key + CASCADE-type dependence; an ordinary FK (eg. .loan.lender_id) is not part of the key.
- PHONES: Separate table PERSON_PHONE (person_id, phone) - one row per number, never "123,456"
- NAME: First_name, last_name. Derived (compute, don't store): loan_days, days_overdue, report_count, last_report_date
- ENTITY vs CODE: collection type and COLLECTION_UNIT have their own data/relations → entities. Role has no data → plain attribute. Artifact value ≠ loan-like valuation (value for that loan, in LOAN_OBJECT)

TASK-7 - RELATIONAL MAPPING.

PERSON (Person_Id PK, first_name, last_name, email UNIQUE)
PERSON_PHONE (Person_Id FK, phone, PK (Person_Id, Phone))
LENDER (Person_Id PK|FK) CURATOR (Person_Id PK|FK, Unit_Id FK) REGISTRAR (Person_Id PK|FK).

ARTIFACT (Artifact_Id PK, catalog_no UNIQUE, title, Unit_Id FK, type_Id FK).

ARTIFACT_OWNERSHIP (Artifact_Id FK, Person_Id FK, Valid_From, Valid_to, Share_Pct, PK (Artifact_Id, Person_Id, Valid_from)).

LOAN (loan_Id PK, lender_Id FK → LENDER, Curator_Id FK, start_date, end_date).

LOAN_OBJECT (loan_Id FK, artifact_Id FK, insured_value, PK (loan_Id, artifact_Id)).

CONDITION_REPORT (report_Id PK, loan_Id, artifact_Id, report_date, notes, FK (loan_Id, artifact_Id) → LOAN_OBJECT)

EXHIBITION (exhibition_Id PK, name) DISPLAY_SLOT (exhibition_Id FK, Slot_no, PK (exhibition_Id, Slot_no))
PLACEMENT (loan_Id, artifact_Id, exhibition_Id, Slot_no, FK → LOAN_OBJECT, FK → DISPLAY_SLOT).

TASK-8- MySQL IMPLEMENTATION

```
CREATE TABLE PERSON (person_id INT AUTO_INCREMENT
PRIMARY KEY, first_name VARCHAR(50) NOT NULL,
last_name VARCHAR (50) NOT NULL, email VARCHAR (100)
UNIQUE);
CREATE TABLE lender (person_id INT PRIMARY KEY, FOREIGN
KEY (person_id) REFERENCES person (person_id);
CREATE TABLE artifact (artifact_id INT AUTO_INCREMENT
PRIMARY KEY, lender_id INT NOT NULL, start_date DATE
NOT NULL, end_date DATE NOT NULL;
  CONSTRAINT chk_dates CHECK (end_date > start_date),
FOREIGN KEY (lender_id) REFERENCES lender (person_id)
  ON DELETE RESTRICT);
CREATE TABLE loan_object (loan_id INT, artifact_id INT,
  insured_value DECIMAL (12,2) CHECK (insured_value >=0),
  PRIMARY KEY (loan_id, artifact_id), FOREIGN KEY (loan_id)
  REFERENCES loan (loan_id) ON DELETE RESTRICT,
  FOREIGN KEY (artifact_id) REFERENCES artifact (artifact_id)
CREATE TABLE exhibition (exhibition_id INT AUTO_INCREMENT
  PRIMARY KEY, name VAR CHAR (100));
CREATE TABLE display_slot (exhibition_id INT, slot_no
  INT, PRIMARY KEY (exhibition_id, slot_no),
    FOREIGN KEY (exhibition_id) REFERENCES exhibition
    (exhibition_id) ON DELETE CASCADE);
```

TASK-9 - VALIDATION QUERIES.

#	Question.	SQL Idea.
Q.1	Loans per lender	SELECT lender_id, COUNT(*) FROM loan GROUP BY lender_id;
Q.2.	Lender name for each loan.	JOIN loan, lender, person.
Q.3.	Artifacts in an exhibition slot.	Join placement, display_slot, artifact.
Q.4.	Artifacts lent more than once.	GROUP BY artifact_id. HAVING COUNT(*) > 1.

TASK-10 - COMPARE MODELS.

MODEL	Person in many loans / queries
File processing	Redundant copies; slow, inconsistent
Hierarchical.	Child has one parent only; curator/lender overlap needs repeated nodes.
Network.	Works, but programmer navigates pointers record by record
Relational (chosen)	Any person in many loans naturally; declarative SQL.

TASK -11- FINAL RECOMMENDATION.

- Choose B. - LOAN entity + LOAN_OBJECT + CONDITION_REPORT.
- Traceability - LOAN_OBJECT → DISPLAY_SLOT → PERSON.
- Tested vs assumed : tested = FK, UNIQUE, date CHECKS.. And query output ; assumed = no overlapping loans, Shares sum 100%. one PERSON for human.
- Limits - Cross-table rules need triggers, duplicate. People with different emails are not detected.

TASK-12 -CHANGED REQUIREMENT- SHARED OWNERSHIP.

DECISION : MODIFY (keep the model, add an ownership-history association).

- A. IMPACT
- B. EVALUATION
- C. ADAPTATION
- D. DEFENSE.

- CONCLUSION- the change touches only ownership. So MODIFY (not redesign) is the smallest correct response.